

Eugene Choi

Contact Information

Mathematics
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Positions

- **Munjeong High School (MHS)**, South Korea
Head, Mathematical Sciences Laboratory 05/2022 -
- **Gachon University (GU)**, South Korea
Research fellow, Statistical and Intelligent Learning (SIL) 02/2026 -

Research Interests

I am interested in **pure mathematics**, particularly in **number theory**, **algebraic geometry**, **algebraic topology**, **probability theory**, as well as various areas of **applied mathematics**. Currently, I am working on a **mathematical framework for analyzing and predicting the behavior of artificially constructed nonlinear complex dynamical systems from a pure mathematical perspective**. My latest research project focuses on the mathematical modeling of overfitting arising from the structural properties of artificial intelligence models, with particular emphasis on developing mathematical approaches to mitigate overfitting and understanding the information distortion that may occur in the process.

Education

- **Munjeong High School**, Seoul, South Korea
Mathematics, 03/2010 - 06/2014
- **Johns Hopkins Center for Talented Youth**, Baltimore, Maryland, United States
02/2017 - 11/2020
- **Cincinnati Gifted and Talented**, MT Healthy, USA
03/2011 - 02/2018

Honors and Awards

- **Euler Prize**, The Korean Mathematical Olympiad, KMO, 2026.
- **Silver Medal**, The Korean Mathematical Olympiad, KMO, 2026.
- **8th Place** in the **The Physics League** competition(On the Experts Section), 2019.
- **Grand Prize**, Chungshin Girls' Middle School Brainy Battle, 2025.
- **Gold Medal**, USA(J)MO, 2020-2021.
- **Honor Roll of Distinction**, American Mathematics Competition 12A,12B, 2020.